

# Gwangmu Lee

Software Engineer @ Swiss National Computing Centre (CSCS)

E-mail: iss300 (at) proton.me    Home: <https://gwangmu.github.io>

## Professional Interest

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### Software Security

Vulnerability Detection & Defense  
Kernel Security

### Software Engineering

Dynamic/Static Software Testing  
Code Analysis & Optimization

### High-performance Computing

System Containerization  
Container Engine Development

## Education

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### Seoul National University (SNU)

Seoul, South Korea.

Ph.D. in Electrical and Computer Engineering (Advisor: Byoungyoung Lee)

Mar 2017 - Feb 2022

### Pohang University of Science and Technology (POSTECH)

Pohang, South Korea.

M.S. in Computer Science and Engineering (Advisor: Jangwoo Kim)

Sep 2014 - Feb 2017

### Pohang University of Science and Technology (POSTECH)

Pohang, South Korea.

B.S. in Physics / Computer Science and Engineering (Double Major)

Mar 2010 - Aug 2014

## Work Experience

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### Swiss National Computing Centre (CSCS)

Lugano, Switzerland.

Software Engineer (Container Engine and Framework)

Apr 2024 - Now

### École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland.

Post-doc Researcher (Software Security), HexHive

Apr 2022 - Mar 2024

### Korea Institute for Advanced Study (KIAS)

Seoul, South Korea.

Research Intern (Astrophysics)

Jul 2013 - Aug 2013

## Publications

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### Architectural Isolation for Sensitive Workloads: Enabling Trusted Research Environments on HPE Cray EX Systems

Cray User Group (CUG), April 2026.

Pim Witlox, Miguel Gila, ..., [Gwangmu Lee](#), ..., and Thomas Schulthess (23 authors)

### Sarus Suite: Cloud-native Containers for HPC

Cray User Group (CUG), April 2026.

Alberto Madonna, Matteo Chesi, [Gwangmu Lee](#), Michele Brambilla, Fawzi R. Mohamed, Felipe A. Cruz.

### **Reflecta: Reflection-based Scalable and Semantic Scripting Language Fuzzing**

The 20th ACM ASIA Conference on Computer and Communications Security (ASIACCS), August 2025.

Chibin Zhang, [Gwangmu Lee](#), Qiang Liu, and Mathias Payer.

### **A Comparative Quality Metric for Untargeted Fuzzing with Logic State Coverage**

arXiv (cs.SE), September 2024.

[Gwangmu Lee](#)

### **SyzRisk: A Change-Pattern-Based Continuous Kernel Regression Fuzzer**

The 19th ACM ASIA Conference on Computer and Communications Security (ASIACCS), July 2024.

[Gwangmu Lee](#), Duo Xu, Solmaz Salimi, Byoungyoung Lee, and Mathias Payer

### **FuzzOrigin: Detecting UXSS vulnerabilities in Browsers through Origin Fuzzing**

The 31st USENIX Security Symposium (USENIX), August 2022.

Sunwoo Kim, Young Min Kim, Jaewon Hur, Suhwan Song, [Gwangmu Lee](#), and Byoungyoung Lee

### **MundoFuzz: Hypervisor Fuzzing with Statistical Coverage Testing and Grammar Inference**

The 31st USENIX Security Symposium (USENIX), August 2022.

Cheolwoo Myung, [Gwangmu Lee](#), and Byoungyoung Lee

### **Constraint-guided Directed Greybox Fuzzing**

The 30th USENIX Security Symposium (USENIX), August 2021.

[Gwangmu Lee](#), Woonchul Shim, and Byoungyoung Lee

### **DynaMix: Dynamic Mobile Device Integration for Efficient Cross-device Resource Sharing**

2018 USENIX Annual Technical Conference (ATC), July 2018.

Dongju Chae, Joonsung Kim, [Gwangmu Lee](#), HanJun Kim, Kyung-Ah Chang, Hyogun Lee, and Jangwoo Kim.

### **Flexon: A Cost-Effective Digital Neuron for Flexible Spiking Neural Network Simulation**

The 45th International Symposium on Computer Architecture (ISCA), June 2018.

Dayeol Lee, [Gwangmu Lee](#), Sunghwa Lee, Dongup Kwon, Youngsok Kim, and Jangwoo Kim.

———— Co-first ————

### **Architecture-aware Automatic Computation Offload for Native Applications**

The 48th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), December 2015.

[Gwangmu Lee](#), Hyunjoon Park, Seonyeong Heo, Kyung-Ah Chang, Hyogun Lee, and HanJun Kim

## Selected Projects

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### **High-Performance Trusted Research Environment**

Jul 2025 - Now

Swiss National Computing Centre (CSCS), System Security Engineer

Keywords: Remote node attestation w/ TPM, Linux security modules, Secure design

### **Investigating Clang Static Analyzer's False-Positives and False-Negatives**

Mar 2023 - Jun 2023

Master's thesis (Cosme Jordan), Supervisor

Keywords: Static code analysis, Student supervision

### **Optimizing Shader Compiler for ARM VLIW-Based Mobile GPU**

Nov 2014 - Jun 2015

Samsung Electronics × High-performance Computing Lab. (HPC), POSTECH, Researcher

Keywords: VLIW compilers, Mobile GPU, Low-level performance optimization

### **POSTECH-KAIST Science War Official AI Game (Diplomatics)**

Jun 2012 - Sep 2012

College game development group (G-POS), POSTECH, Lead Programmer

Keywords: P2P game development

## Recognition

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### 26th Graduate Student Scholarship

Korea Foundation for Advanced Study (KFAS)

Mar 2015 - Feb 2020

### National Science and Engineering Scholarship

Korea Student Aid Foundation (KOSAF)

Mar 2010 - Feb 2014

## Teaching Experience

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|--------------------------------------------------------------------------------------|-------------|
| <b>Introduction to Programming</b> , Dongji High School, Instructor                  | Winter 2014 |
| <b>Introduction to Programming</b> , Dongji High School, Instructor                  | Summer 2015 |
| <b>Object-Oriented Programming</b> , POSTECH, Teaching Assistant                     | Fall 2015   |
| <b>Computer Architecture</b> , POSTECH, Teaching Assistant                           | Fall 2016   |
| <b>Computer Organization</b> , Seoul National University, Teaching Assistant         | Spring 2017 |
| <b>Digital Logic Design</b> , Seoul National University, Teaching Assistant          | Fall 2017   |
| <b>Cyber Security and Blockchain</b> , Seoul National University, Teaching Assistant | Fall 2019   |

## Professional Services

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|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <i>Invited Talk</i>    | <b>The Chronicle of Software Vulnerability Detection</b><br>Kyung Hee University. Slides available at <a href="https://gwangmu.github.io/experience/">https://gwangmu.github.io/experience/</a> . | Jun 2023     |
| <i>Reviewer</i>        | <b>IEEE Transactions on Dependable and Secure Computing (TDSC)</b><br><b>IEEE Transactions on Software Engineering (TSE)</b>                                                                      | 2023<br>2023 |
| <i>Prog. Committee</i> | <b>Intl. Symposium on Research in Attacks, Intrusions and Defense (RAID)</b><br><b>Intl. Symposium on Research in Attacks, Intrusions and Defense (RAID)</b>                                      | 2024<br>2026 |

## Skills & Etc

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| <i>Language</i>                      | <b>Korean</b> (native), <b>English</b> (semi-fluent), <b>Italian</b> (baby)                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <i>Programming Language</i>          | Day-to-Day: <b>C</b> , <b>Python</b> , <b>Bash</b><br>Confident: <b>Go</b> , <b>Rust</b> , <b>C++</b><br>Familiar: <b>Java</b> , <b>Scala</b> , <b>OCaml</b> , <b>Verilog</b>                                                                                                                                                                                                                                                                                                                                                            |
| <i>Personal Proj.<br/>(Selected)</i> | <b>Project SoftBounded: Memory Safety for Ultra-Low-Level Applications</b><br>Engineering project to make SoftBound [PLDI09] practical for ultra-low-level software, <a href="#">link</a> .<br><br><b>English-to-Hangul Transliterator</b><br>Mini-project that transliterates English sentences to augmented Hangul sentences, <a href="#">link</a> .<br><br><b>Squashfused (Sqaushfuse-dee)</b><br>Mounting SquashFS inside an unprivileged rootless container, <a href="#">link</a> , <a href="#">blog1</a> , <a href="#">blog2</a> . |